Stable and verifiable latent dynamics for world models

[TODO: Author Name(s)]
[TODO: Affiliation, Department, Institution, City, Country]
[TODO: corresponding author email]

Abstract

World models of the Dreamer family optimize policies over latent trajectories imagined by a recurrent state-space model, yet nothing in their training constrains those latent dynamics to be stable, and the categorical latent output blocks direct formal certification. We report two measured obstacles and an evaluation standard for the resulting problem. First, a sampling-to-proof gap audit of the one-step latent map under a frozen actor: a Lyapunov decrease condition, a large random-sampling baseline, and branch-and-bound verification with a reachability gate. Second, a measured verifier wall: interval bound propagation asserts on the categorical output at any useful width, so every branch-and-bound box is recorded as *unknown* with its reason, never as a violation. Third, a five-mode failure atlas from a controlled latent-space study shows why naive stability constraints fail, from which we derive design rules and an aliveness-gated evaluation standard. All claims name the object, region, and condition that produced them.

1 Introduction

Model-based reinforcement learning via world models has become a general approach to control. DreamerV3 [1, 2] masters over 150 tasks from a single configuration, and its policy is optimized over trajectories *imagined in latent space* by a Recurrent State-Space Model (RSSM) [3] – never in the environment. This is precisely why the stability of the latent dynamics matters: if the imagined trajectory drifts off-distribution, diverges, or collapses to a spurious fixed point, the policy is optimized against a model of a world that does not exist. DreamerV3 is trained with no constraint that the latent transition map be contractive, energy-bounded, or otherwise stable, and its categorical latent output complicates any formal argument about the imagined dynamics.

Two lines of work have begun to address this. The first constrains the *learned* dynamics so imagination stays well behaved; the most recent example is Koopman Dreamer [4], which replaces the deterministic latent core with spectrally constrained rotation–scaling blocks of bounded radius and derives a multi-step rollout-error bound. The second verifies a *frozen* model after the fact with neural Lyapunov methods [7, 8] and sound bound propagation [5, 6]. Both lines face a common, largely unexamined question: how large is the gap between what random sampling finds and what a verifier can prove, and can scalar or spectral constraints actually shape the stability of a learned latent map?

This Letter reports three measured results on that question. (i) A sampling-to-proof gap audit for the one-step latent map of a DreamerV3-size model under a frozen actor. (ii) A structural verifier wall: interval bound propagation cannot bind the categorical latent output at any useful width. (iii) A five-mode failure atlas, from a controlled latent-space study, of how the optimizer routes around scalar stability constraints – including the decoupling of spectrum-sum constraints from the leading Lyapunov exponent and the decoupling of finite-time spectral proxies from the asymptotic exponent. From the atlas we derive design rules for stabilization attempts and an evaluation standard that gates attractor-overlap metrics on rollout aliveness.

Honest framing. This is a methods and measurement Letter. The verifier wall and the failure atlas are measured; the direct application to a *trained* DreamerV3 on a control task is [TODO: pending GPU audit], and no quantitative claim about DreamerV3 itself is made until that audit lands. The latent-space study uses Lorenz-63, which is not a natural GENERIC system; its learned energy and dissipation are proxies. We claim on-attractor stability and invariant recovery, not recovery of true thermodynamics.

2 The one-step latent map and the sampling-to-proof gap audit

DreamerV3's world model factorizes the transition into a deterministic recurrent core and a stochastic posterior over discrete classes: $d_t = f_\theta(d_{t-1}, s_{t-1}, a_{t-1})$ and $s_t \sim q_\phi(s_t \mid d_t, o_t)$, with a prior $p_\phi(s_t \mid d_t)$ used at imagination time. The object we audit is the one-step latent map under a frozen deterministic actor π,

$$\boldsymbol{z} = (d, s) \;\mapsto\; \boldsymbol{z}' = \big(d',\ \mathrm{softmax}(\mathrm{logits}')\big), \qquad d' = \mathrm{core}\big(d, \pi(\boldsymbol{z})\big), \tag{1}$$

where the stochastic output is replaced by the *mean* of the prior categorical, and $\pi(\boldsymbol{z}) = \tanh(\mathrm{mean}(\mathrm{actor}))$. The model is audited as trained; only the Lyapunov candidate V moves. Nothing about the imagined multi-step rollout is certified, and nothing larger than the size-1M configuration (deter 512, hidden 64, stoch 32×4) is considered.

The audit (Fig. 1) verifies the Lyapunov decrease condition

$$\mathrm{cond}(\boldsymbol{z}) = V(\boldsymbol{z}) - V(\boldsymbol{z}') > 0 \tag{2}$$

over a box region built *from* the model's own latent states: a frozen checkpoint is rolled out on-policy, the posterior latents are recorded, and the certified region is the coordinate box around them. The region is therefore on-distribution by construction; the certified map is the prior-mean surrogate, and the distinction is recorded in the artifact, never conflated.

Sampling draws $\boldsymbol{z}$ uniformly from the box and evaluates Eq. (2) exactly. Branch-and-bound uses CROWN-style relaxations [5] as implemented in auto_LiRPA [6], with the verifier source pinned by the SHA-256 of the driver. A verdict is one of three: *violation* (a counterexample the model demonstrably reaches), *certified* (the bound proves the condition on the whole box), or *unknown* (bounds stayed loose, no counterexample found). *Unknown* is verifier incompleteness: it is never evidence of safety, and it is never a gap and never a partial finding. The branch-and-bound region is an annulus around the latent fixed point $\boldsymbol{z}^*$, holed along the k coordinate dimensions with the largest halfwidths so the number of boxes scales with k, not with the latent dimension; the fixed point itself, where $\mathrm{cond}(\boldsymbol{z}^*) = 0$ by construction, is excluded.

3 The verifier wall on categorical latents

The certified graph contains the categorical output $s' = \mathrm{softmax}(\mathrm{logits}')$, i.e. an $\exp / \sum \exp$ with a division by the relaxed sum. Interval bound propagation asserts on this node whenever the relaxed logits interval spans a region where the softmax denominator can be non-positive in the bound computation; at the measured vacuity of the intermediate bounds this happens on the first box ("AssertionError in BoundReciprocal"), and the same structural failure repeats identically on every box. Because the bound-graph build dominates the per-box cost (measured: $\sim$4 minutes for the graph trace on a size-1M graph on CPU), running branch-and-bound without the wall gate wastes hours to rediscover the same assertion.

The audit therefore probes the wall once, then records every box as *unknown* with the reason "categorical output not interval-bindable at this width". This is the toolchain boundary, now *measured* rather than assumed. The sampling baseline is the empirical result, and a sampling violation is the only thing that can become a finding; it must survive the reachability gate first.

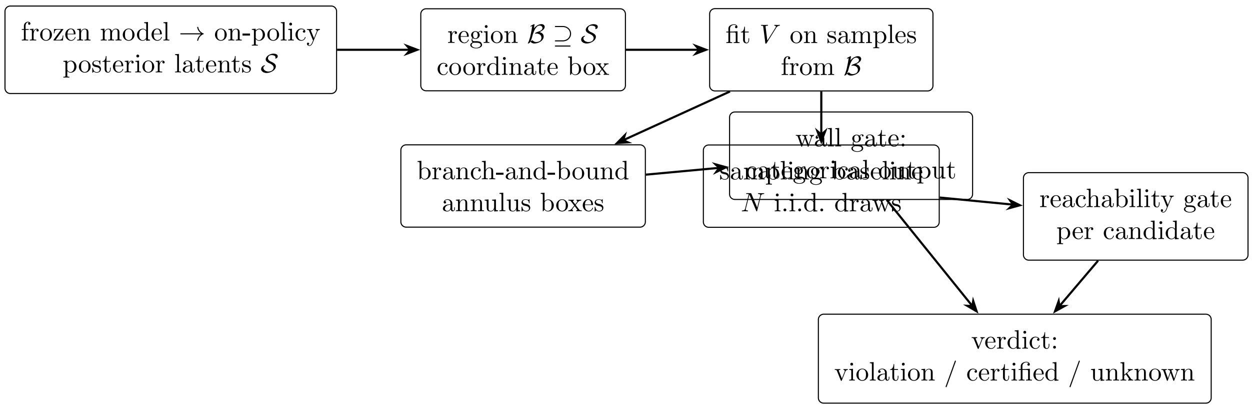

Figure 1: The sampling-to-proof gap audit. The region is built from the model's own on-policy latents; sampling and branch-and-bound compete on the same decrease condition; candidates pass a reachability gate; a structural wall probe records boxes that the verifier cannot bind.

Table 1: Measured audit record on the smoke configuration (weight-initialized surrogate; CPU; not a trained model).

Quantity	Measured
Latent fixed point residual	2.98×10^{-8} (converged)
Region coverage of steady-state latents	83.8%
Sampling baseline	0 / 12 866 violations
Worst sampling condition	$+2.2 \times 10^{-3}$
Branch-and-bound boxes	4 (k-dim hole)
Wall record	every box *unknown*: categorical output
Verdict	`gap_demonstrated` = `false`

On the smoke configuration – a weight-initialized surrogate, *not* a trained model – the full pipeline runs end to end on CPU (Table 1). *None of these numbers describe a trained model*; they establish that the pipeline is sound and reproducible, which is their only claim. The trained-model audit on dm_control cartpole_balance is [TODO: pending GPU]; on the categorical wall its expected verdict is *unknown*, which is verifier incompleteness and not a finding.

4 The failure atlas of latent-space stabilization

The wall says the frozen model cannot be certified directly. The remaining question is whether it can be *stabilized* – and here we report the atlas from a controlled study: a Hamiltonian-JEPA predictor trained end to end on chaotic Lorenz-63 under a nonlinear coordinate warp, in five ablation arms with matched data and steps (unconstrained MLP; rigid Hamiltonian; naive metriplectic; fixed divergence-free-J metriplectic with invariant dissipation gate; and the fixed arm plus a differentiable QR spectral alignment). The study is reproducible on CPU with fixed seeds. The atlas is the *negative result*: imposing physical or thermodynamic constraints inside a self-supervised latent space does not, by itself, yield physical dynamics, because the optimizer routes around each scalar constraint through a geometric exploit (Table 2).

(1) Freeze. Multi-step alignment is minimized by a predictor that does not move: compounding K-step error makes motion locally costlier than none, and the field gradients collapse. A horizon

Table 2: Documented failure modes of latent-space stabilization (Lorenz-63, nonlinear warp; signatures as documented, CPU reproduction in progress at draft time).

#	Mode	Measured signature
1	Representation freeze	motion $\sim 10^{-3}$ vs target ~ 0.4; penalty ~ 0
2	Scale-conditioning paradox	proxy spread ~ 0.036 vs $\|\partial H/\partial p\| \sim 3.2$
3	Skew loophole + dead R saddle	$\mathrm{tr}(R)/n = 0.00000$ in every naive run
4	Sum-vs-λ_1 decoupling	contraction -12.8, λ_1 blows up to $+3.5$ (true ~ 0.88)
5	Finite-time spectral trap	short-horizon $\hat{\lambda}_1 \to -0.17$, true 200-step $+1.44$

curriculum and a detached motion-magnitude penalty are partial mitigations.

(2) Scale-conditioning paradox. Matching the raw magnitude of a kinetic-energy proxy clamps the dynamics: the proxy's spread is $\sim 30\times$ below what the integrator needs, so magnitude matching freezes the model. Comparing *z-scores* of both sides – keeping only the shape/correlation constraint – restores an $O(1)$ loss and frees the scale.

(3) Skew loophole and dead R saddle. A state-dependent skew $\mathbf{J}(\boldsymbol{z})$ is not divergence-free, so it can supply demanded volume contraction with no dissipation at all, leaving $\mathbf{R}$ off; and $\mathbf{R} = \mathbf{L}\mathbf{L}^\top$ with $\mathbf{L}$ zero-initialized has $\partial\mathbf{R}/\partial\mathbf{L} = 0$ at $\mathbf{L} = 0$, a dead saddle no gradient can lift. Both must be closed – fixed constant canonical skew, small nonzero $\mathbf{L}$ init – before dissipation engages ($\mathrm{tr}(R)/n \sim 5.6$; contraction -13.667).

(4) Sum-vs-λ_1 decoupling. Pinning the divergence pins the *sum* of the Lyapunov spectrum, but the sum does not pin any individual exponent. The fixed arm reaches strong contraction while λ_1 blows up to $+3.5$ (true ~ 0.88): the negative exponents compensate. Confirmed four times ($\lambda_1 \in \{-0.42, +1.44, +3.5, +4.29\}$, all with the "correct" sum). A scalar volume constraint is structurally incapable of shaping the leading exponent – the direct caution for spectral designs that constrain aggregate spectrum information, such as bounded-radius backbones [4].

(5) Finite-time spectral trap. At short horizons the finite-time QR exponent decouples from the asymptotic λ_1: training drives $\hat{\lambda}_1$ to -0.17 while the true 200-step exponent is $+1.44$ (opposite sign). At long horizons the proxy is trainable but nothing constrains global boundedness: the evaluation rollout diverges to $+\infty$. The trap moves from "wrong proxy" to "right proxy, unbounded global trajectory".

The overlap trap (evaluation standard). Chamfer/Hausdorff distance to the encoded attractor *rewards* a frozen predictor: a stationary point inside the attractor cloud scores near-perfect while being dynamically dead. Geometric overlap must be gated on the rollout being alive and chaotic ($\lambda_1 > 0$ and a sane motion ratio), or replaced by a motion-invariant statistic.

Transfer caveat. The atlas is measured on Lorenz-63 latent dynamics, not on DreamerV3's RSSM latents. Its role here is to make the failure modes concrete and checkable; the audit of Section 2 is the tool that tests whether the same exploits appear in a trained world model's latent space. That test is the primary open item.

5 Design rules and the evaluation standard

From the wall and the atlas we derive the following rules; each is stated with the failure mode it is designed to avoid (Table 3).

1. **Certify a verifiable surrogate, not the categorical output.** The categorical output is not interval-bindable at useful width. Certification must target the deterministic component

Table 3: Design rules for stabilizing latent dynamics, indexed by the failure mode each avoids.

Rule	Avoids
Certify a verifiable surrogate	categorical wall (Section 3)
Constrain modes, not sums	failure 4 (sum-vs-λ_1)
Break dead saddles with init	failure 3 (dead R saddle)
Shape constraints, not magnitudes	failure 2 (scale paradox)
Curriculize horizon; gate on aliveness	failures 1 and 5; overlap trap
Report unknown honestly	overstatement

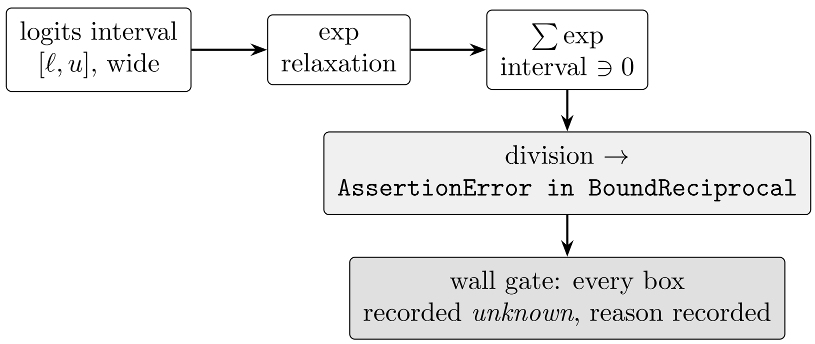

Figure 2: The measured categorical wall. The softmax / sum-division node asserts in the interval pass at any useful width, so the audit probes once and records every box as *unknown* with its reason.

of the latent map, or a structurally verifiable surrogate, with the stochastic output treated as the documented boundary of the toolchain.

2. **Constrain modes, not sums.** A scalar volume/spectrum-sum constraint cannot shape the leading Lyapunov exponent. Spectral designs should constrain per-mode radii and verify the *leading* exponent directly, not the aggregate.

3. **Break dead saddles with principled initialization.** Factorized dissipation or learned structure that starts at zero and is penalized toward engagement needs a nonzero initialization and a gradient-path check.

4. **Use shape constraints, not raw magnitudes.** Scale-free (z-score/correlation) anchors decouple magnitude from shape and prevent scale-starvation.

5. **Curriculize the horizon and gate evaluation on aliveness.** Short-horizon objectives freeze the predictor, and finite-time proxies decouple from asymptotic exponents; evaluation must include a long-horizon rollout, an aliveness gate, and a leading-exponent check, and must never report attractor overlap without it.

6. **Report the gap honestly.** *Unknown* verdicts, loose bounds, and walls are verifier incompleteness: never evidence of safety, never a gap. Only a reachable violation counts, and it must survive a reachability gate.

6 Discussion

What is measured. The verifier wall is a measured property of auto_LiRPA 0.7.2 on the categorical output, reproducible on CPU in minutes. The failure atlas is a measured, reproducible negative result on a controlled latent-space benchmark. The audit pipeline is measured end to end on a surrogate.

What is not yet measured. A trained DreamerV3 model's latent dynamics have not yet been audited; that audit requires a GPU training run and is the primary open item. The transfer of the failure modes from Lorenz latents to RSSM latents is a hypothesis the audit is designed to test, not a claim. *Unknown* verdicts on the categorical wall are expected and are not findings.

Scope. The certified object is the one-step prior-mean map under a frozen policy; nothing about imagined multi-step rollouts or the stochastic posterior is certified. The region is built from on-policy posterior latents; the certified map is the prior-mean surrogate.

Implications for the field. World-model-based control deployed on physical systems needs latent trajectories that stay bounded and on-distribution over planning horizons, and a way to argue about that formally. This Letter contributes the measurement machinery (the gap audit), the obstacle (the categorical wall), and the cautionary evidence (the failure atlas) that any stabilization design must confront. We do not claim to know any team's internal priorities; we claim only that these are the questions any stability argument for a Dreamer-style latent space must answer.

7 Methods

Audit procedure. The pipeline (Fig. 1) is: fit the Lyapunov candidate V by gradient descent on samples from the region (penalizing violations of Eq. (2)); run the sampling baseline (N i.i.d. draws, worst condition and count recorded); run branch-and-bound over the annulus boxes with a per-box budget; apply the reachability gate to any candidate violation by simulating the model's closed-loop dynamics toward it; probe the categorical wall once and record unbindable boxes as *unknown* with the reason. The verdict is `gap_demonstrated` only when sampling found nothing and branch-and-bound found a reachable violation.

Region construction. A frozen checkpoint is rolled out on-policy; posterior latents are recorded; the region is the coordinate box around them. The annulus excludes the latent fixed point and is holed along the k coordinate dimensions with the largest halfwidths.

Verifier pinning. The verifier is auto_LiRPA 0.7.2 at GitHub commit 5a098e8f (the release the project was validated against; PyPI releases carry incompatible legacy torch constraints). The driver that produced a result is pinned by the SHA-256 of its source. The categorical wall is probed by a single structural test of the interval pass on the softmax/sum-division node.

Latent-space benchmark. Lorenz-63 ($\sigma = 10$, $\rho = 28$, $\beta = 8/3$) is sampled at $\mathrm{d}t = 0.01$, lifted by a fixed nonlinear diffeomorphism into a 64-D latent space, and predicted by five JEPA predictor arms (unconstrained MLP; rigid Hamiltonian; naive metriplectic; fixed-J metriplectic with invariant dissipation gate; fixed arm plus differentiable QR spectral alignment), each trained with matched data, steps, and seeds. Evaluation is a 200-step rollout judged by boundedness, leading Lyapunov exponent (Benettin), phase-space contraction rate, attractor overlap (Hausdorff/Chamfer), and dissipation engagement $\mathrm{tr}(R)/n$. The empirical true exponent at $\mathrm{d}t = 0.01$ is $+0.884$ (textbook 0.9056); the true divergence is -13.6667. All reported numbers come from fixed-seed runs; per-seed spread is reported where available.

Code and data availability. The audit pipeline, the port of the DreamerV3 size-1M networks, the notebook builder, and this manuscript are in the public repository github.com/sehajr-singhs/rl-wm-audit. The failure atlas benchmark, results, and figures are in github.com/sehajr-singhs/spectral-topological-decoupling. All code runs on CPU.

Data and Code Availability

See Methods.

Author Contributions

[TODO]

Competing Interests

The authors declare no competing interests.

Acknowledgements

[TODO]

References

[1] D. Hafner, J. Pasukonis, J. Ba, and T. Lillicrap, "Mastering diverse control tasks through world models," *Nature*, vol. [TODO], 2025. [TODO: verify volume/pages/DOI.]

[2] D. Hafner, J. Pasukonis, J. Ba, and T. Lillicrap, "Mastering diverse domains through world models," arXiv preprint arXiv:2301.04104, 2023.

[3] D. Hafner, T. Lillicrap, I. Fischer, R. Villegas, D. Ha, and H. Lee, "Learning latent dynamics for planning from pixels," in *Proc. ICML*, 2019.

[4] J. Li, X. Zhang, H. Xie, Y. Lan, W. Pan, and X. Xu, "Koopman Dreamer: Spectrally constrained latent dynamics for stable world-model imagination," arXiv preprint arXiv:2607.19719, 2026.

[5] H. Zhang, T.-W. Weng, P.-Y. Chen, C.-J. Hsieh, and L. Daniel, "Efficient neural network robustness certification with general activation functions," in *Proc. NeurIPS*, 2018.

[6] K. Xu, *et al.*, "auto_LiRPA: Automatic linear relaxation based perturbation analysis for neural networks," arXiv preprint arXiv:2002.12920, 2020.

[7] Y.-C. Chang, N. Roohi, and S. Gao, "Neural Lyapunov control," in *Proc. NeurIPS*, 2019.

[8] C. Dawson, S. Gao, and C. Fan, "Safe control with learned certificates: A survey of neural Lyapunov, barrier, and contraction methods," arXiv preprint arXiv:2202.11762, 2023.

[9] Y. LeCun, "A path towards autonomous machine intelligence," Open Review preprint, 2022.

[10] M. Assran, *et al.*, "Self-supervised learning from images with a joint-embedding predictive architecture," in *Proc. CVPR*, 2023.

[11] L. Maes, Q. Le Lidec, D. Scieur, Y. LeCun, and R. Balestriero, "LeWorldModel: Stable end-to-end joint-embedding predictive architecture from pixels," arXiv preprint arXiv:2603.19312, 2026.

[12] J. Miller and M. Hardt, "Stable recurrent models," in *Proc. ICLR*, 2019.

Draft status. Authors TODO; references to verify; the trained-model audit [TODO: GPU run]; the CPU reproduction of the atlas table in progress at draft time. Nothing in this draft has been submitted or peer reviewed.